

Procedures Lab: Instructing NAO6 to Complete 3 Pushups

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The purpose of this document is to evaluate and visualize the process behind communicating steps to complete an action in a specific, concise, and clear way to a human or humanoid being, in this case more specifically, Bob's cousin, the NAO6 robot, who may not understand how to complete the action without guidance. I make use of flowcharts and a numbered procedures layout that NAO6 could use to conduct three pushups from standing position and return back to a standing position once completed. This has great importance in computer science as it demonstrates that instructing computers to complete a task usually involves and requires deep thought. Terms for certain topics are listed within the glossary section.

Precautions

- Ensure that NAO6's environment has no traffic, is rigid, flat, free of any holes, solid, has at least enough friction similar to wood, and is at least 10 feet in width and length to ensure sufficient space for NAO6 to complete pushups.

CAUTION: These are recommended requirements, NAO6 can be placed on other surfaces, but unsuccessful completion of pushups can result if a similar environment is not used.

Prerequisites

1. Place NAO6 in a suitable flat and level environment.

DANGER: If NAO6 is not placed in a stable, long and wide, flat, and level environment, damage to NAO6 can ensue.

2. Instruct NAO6 that you are about to define steps to follow on how to conduct 3 pushups.

NOTE: NAO6 must be called by name and waited on to respond prior to being assigned a task.

Procedures

Get Into Formation

1. Rotate both knee joints fully out while lowering hip joint and remaining upright
2. Rotate wrists until rear of hands face the floor.
3. Extend shoulder and elbow joints outwards and forward all the way
4. Rotate upper torso down from upright position until wrists touch ground surface

CAUTION: NAO6 may fall during this transition, so having someone prepared to catch NAO6 during this transition period is highly recommended to prevent damage.

5. Rotate both knee joints outwards while keeping shoulder and elbow joints stretched

Conduct Pushup

6. Rotate elbow joints inwards
7. Rotate elbow joints outwards

Conduct Pushup

8. Rotate elbow joints inwards
9. Rotate elbow joints outwards

Conduct Pushup

10. Rotate elbow joints inwards
11. Rotate elbow joints outwards

Return to Standing Position

12. Bend both shoulder joints outwards
13. Bend both elbows' joints inwards
14. Extend both arms outwards until knuckles touch bottom surface
15. Rotate knees inwards
16. Rotate hips until toes touch the bottom surface

CAUTION: NAO6 may fall during this transition, so having someone prepared to catch NAO6 during this transition period is highly recommended to prevent damage.

17. Rotate hip joint fully right
18. Extend left knee joint outwards
19. Rotate left shoulder joint inwards
20. Rotate hip joint fully left
21. Extend right knee outwards
22. Rotate upper torso back slowly and rotate elbow out at same time
23. Bend knees in until upright

Post-performance Activities

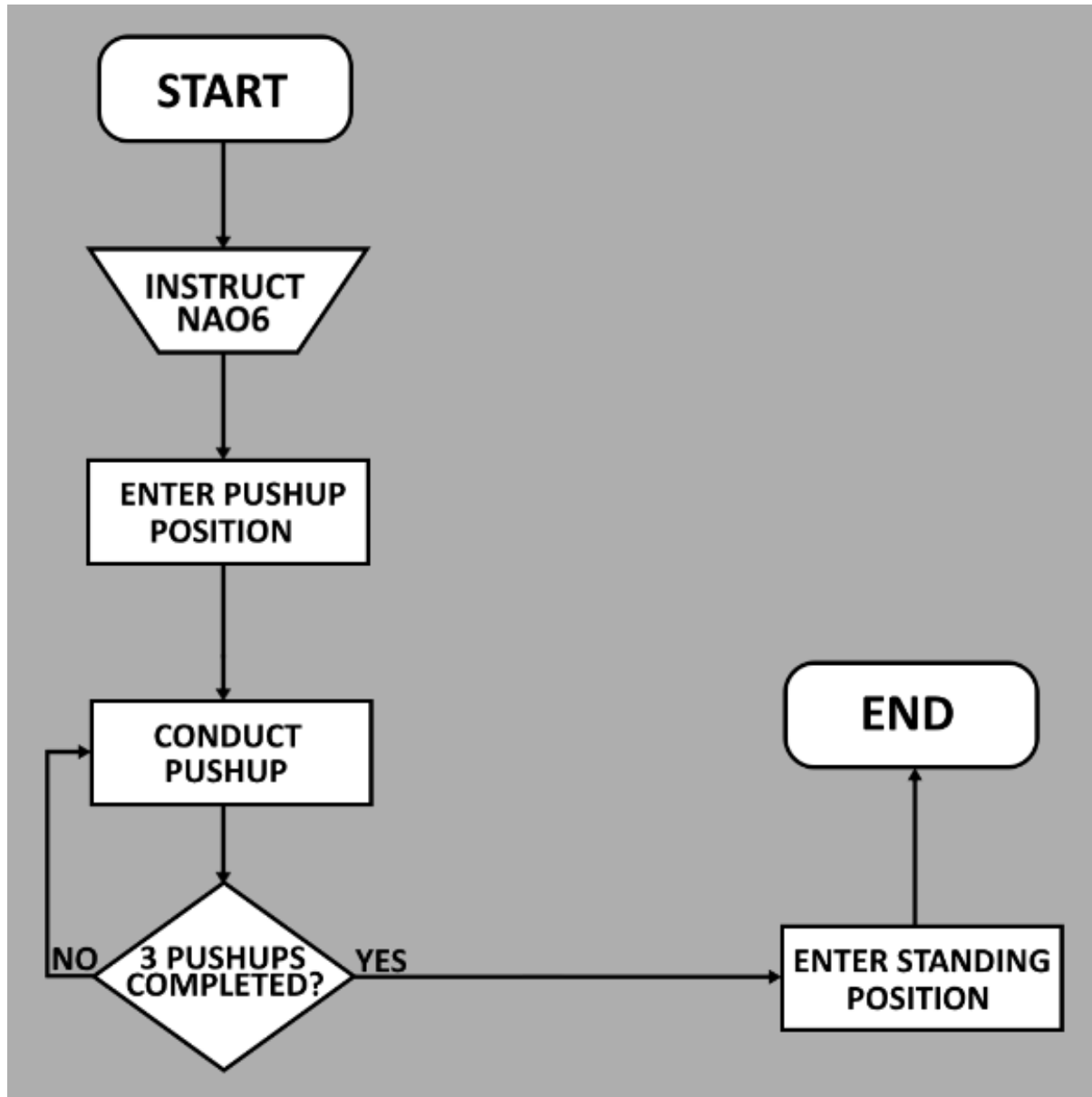
Following NAO6's completion of the pushups, some things I realized could help improve the process of describing instructions procedures to a robot by analyzing and identifying areas for improvement on instructing a humanoid to complete a task can be improved by using some of the following:

- Testing and documenting results more than once to find similarities and differences.
- Keeping track of every small movement done by an example used as a reference.

Flow Chart

Figure 1

NAO6's 3 Push-Ups Process Visualized



Note. ISO flowchart of a basic overview of the process of NAO6 completing the task of doing 3 pushups.

Glossary

Term	Definition
Bob	Robot at Central Washington University that has the ability to conduct push-ups.
Caution	Possible damage to equipment or data; or potential problems in outcome of action.
Danger	Warning about possible serious or fatal injury to self or others.
NAO6	Robot built exactly like Bob, but does not have the built in pre-instructed ability to conduct push-ups.
Note	Reminder or minor problem in outcome of doing something.
Warning	Warning about possible minor injury to self or other.

References

POBJuice. (2019). *Human Interaction with The NAO Robot*. YouTube.

<https://www.youtube.com/watch?v=5IrOXU6Idbw>

Salter, R. (2026). *Procedures* [PowerPoint slides]. Central Washington University.